



Regular Research Article

Electroconvulsive Therapy in the Oldest-Old Patients With Depression: Response and Remission Rates, Prognostic Factors, Adverse Events and Mortality

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ABSTRACT

Objectives: Depression is common in the oldest-old population (aged ≥ 85 years), a population in which research is lacking and the effects of psychopharmacological treatment may be adverse. Electroconvulsive therapy (ECT) appears to be well tolerated by older patients with depression; however, it is used rarely in clinical practice and research concerning the oldest-old is lacking. The objective of this study was to assess response and remission rates, as well as adverse effects, in oldest-old depressed patients treated with ECT. **Methods:** This was a nationwide Swedish register study including 522 oldest-old patients treated with ECT for depression. Two propensity score matched control groups were also included: young patients treated with ECT, and oldest-old depressed patients not treated with ECT. Response and remission rates, clinician-reported adverse events, prognostic factors (including comorbid diseases and treatment parameters), adverse events requiring hospitalization, as well as deaths within 1 week from discharge, were assessed. **Results:** The oldest-old receiving ECT reported higher response (81.2%) and remission (53.3%) rates than younger counterparts (67.4% versus 27.4%, respectively). Severe psychotic depression was the only prognostic factor for response and remission. The oldest-old experienced fewer adverse events, most commonly confusion and cardiovascular

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*complications, compared to controls. Compared with oldest-old patients not receiving ECT, the ECT group had fewer hospitalizations following discharge. **Conclusion:** ECT is a viable treatment option for oldest-old patients with depression, who show higher response and remission rates than younger patients, as well as fewer hospitalizations than oldest-old patients not treated with ECT. (Am J Geriatr Psychiatry 2025; 33:1065–1076)*

Highlights

- **What is the primary question addressed by this study?**

This study aimed to assess response rates and side effects of electroconvulsive therapy (ECT) in the treatment of oldest-old patients, aged 85 or older, suffering from depression.

- **What is the main finding of this study?**

The study found that the oldest-old patients had high response rates to ECT, significantly higher than a younger control group, and experienced fewer deaths and adverse events requiring hospitalization compared to depressed oldest-old patients who did not receive ECT. Additionally, the oldest-old patients reported significantly fewer adverse events from ECT than their younger counterparts who received ECT.

- **What is the meaning of the finding?**

The results suggest that ECT could be a viable treatment for depression in the oldest-old.

INTRODUCTION

Depression is common, and associated with an elevated risk of morbidity and mortality.¹ On a global level, approximately 270 million people suffer from major depressive disorder (MDD), and the prevalence increases with increasing age.² Severe MDD afflicts about 2% of people aged 55 years and older, and the incidence continues to increase with advancing age,³ despite commonly being underdiagnosed in this population.⁴ Depression among old people is also associated with a higher incidence of suicide when compared with younger counterparts,⁵ although suicide rates in those aged 70 or more has fallen considerably in more recent years.⁶ The population aged 85 years or older is often referred to as the oldest-old. As global life-expectancy continues to increase, this population is growing rapidly, while remaining an understudied age group.⁷

Psychopharmaceutical interventions are employed most commonly for management of depression among the oldest-old patients. Unfortunately, there is a low confidence in such treatment paradigms in the oldest-old, due partly to limited scientific evidence⁸ and concerns of polypharmacy-related side effects,⁹

as well as a generally heightened risk of adverse events compared with younger counterparts.¹⁰ On a similar note, multimorbidity, the co-occurrence of at least two chronic somatic illnesses, is common in the oldest-old,¹¹ one reason being age-related systemic inflammation, or “Inflammageing”.¹² As a result, the oldest-old psychiatric patients often take several medications to treat diseases, increasing the risk of polypharmacy and negative drug-drug interactions¹¹; common combinations include NSAIDs for arthritis and antidepressant SSRIs, which can cause gastrointestinal bleeding.¹³

Consequently, there is a clear need for alternative treatment modalities for the oldest-old patients with severe depression. Electroconvulsive therapy (ECT), used to treat severe depression, involves triggering of a generalized epileptic seizure during muscle relaxation and anesthesia, with the seizure subsiding naturally after a duration of 30–60 seconds.¹⁴ Importantly, ECT is a noninvasive antidepressant treatment that is well tolerated by old people. Previous literature reveals a spectrum of findings with respect to the efficacy of ECT among old patients. The majority of studies show good results in old patients,^{15,16} including a recent study of 46 patients aged 80 years

or older,¹⁷ while others have indicated that age has no clinical relevance regarding the outcome of treatment.¹⁸ It is noteworthy that several studies demonstrated substantial improvements in patients, with response rates ranging from 70% to over 90%, irrespective of age, following ECT for depression.¹⁹⁻²¹ These findings are compelling, although they are limited by small sample sizes. Also, utilization of ECT remains relatively limited in the context of oldest-old patient care on a global level. The reasons for this are believed to be multifactorial, and involve stigma and the risks of cognitive side effects.²² Importantly, it remains uncertain whether the outcomes of ECT in oldest-old patients are affected by comorbid somatic conditions, or multimorbidity.

Taken together, the data suggest that more studies are needed regarding both the efficacy and safety of ECT as a treatment for older patients with depression. Above all, there is a scarcity of research concerning ECT for depression among the oldest-old patients. This study leveraged national Swedish registers and aimed to 1) assess response rates and 2) adverse events of ECT as a treatment for the oldest-old patients suffering from depression, and to 3) assess the potential influence of comorbid somatic conditions on these outcomes. We also compared treatment outcomes with those of two propensity score matched groups: patients aged 18–35 who also underwent ECT for depression, and patients aged 85 or older who were inpatient treated for depression but who did not receive ECT.

MATERIAL AND METHODS

Study Design

This study was a register-based study based on nationwide population registers from Sweden. Data were collected from the Swedish National Quality Register for ECT (Q-ECT),²³ the Swedish National Patient Register (NPR),²⁴ and the Swedish National Cause of Death Register (CDR).²⁵ The data from Q-ECT were used to examine the proportion of patients aged ≥ 85 years who improved after treatment with ECT, according to Clinical Global Impression Improvement (CGI-I).²⁶ Only patients with a rating on CGI-I were included in the study. Two propensity score matched control groups were used. One of these

comprised individuals aged 18–35 years from the Q-ECT, and was used to mitigate confounding factors while providing a comparison group characterized by a relative absence of multimorbidity. This particular age span was chosen because it includes adults before middle age, which is often defined as starting from age 35.²⁷ In this age group, somatic comorbidities are expected to be less prevalent. A second control group comprised patients aged ≥ 85 years from the NPR, who received inpatient treatment for depression, but did not receive ECT. Although the NPR contains less detailed information than the Q-ECT, this group allowed assessment of whether adverse outcomes following depression treatment could be related to ECT, or if they were simply common in oldest-old patients with depression.

Data Sources

Data pertaining to ECT treatment was sourced from the Q-ECT registry, which is a national register that collects information regarding ECT in Sweden. This comprehensive registry encompasses all hospitals across the country that offer ECT services, boasting a coverage rate of 92% in 2020.^{23,28} In Sweden, ECT is available at nearly all hospitals with psychiatric inpatient wards (total, 45 units in 2023), and all report to the register. The ECT units treat patients regardless of age, but some hospitals have psycho-geriatric wards. Usually, three ECTs are given per week during index-series, but sometimes only two treatments/week are preferred. The treatments are dominated by right unilateral (RUL) electrode placement (approximately 90% of all treatments) and age- and sex-based dosage. The seizure threshold method is used rarely in Sweden. The machines used are either Thymatron or MECTA, with a maximum 1,100 mC maximum charge. Note that the maximum setting available for these machines is double that in some other countries, including the United States. Patients are sedated with thiopental or propofol and muscle relaxed with succinylcholine; EEG-monitoring is standard.²⁹

Complementary data concerning comorbid conditions were extracted from the Swedish NPR, a database that houses records encompassing diseases and treatments administered in specialized healthcare settings in Sweden.²⁴ This register was also used to identify a comparison group of oldest-old patients

who were treated for depression but not with ECT, as well as hospitalizations during or within 7 days from treatment. The dataset spanned the years 1998–2022.

Information about death following ECT was collected from the Swedish national CDR, which contains information about all deceased persons, as determined by a physician, in Sweden since 1961.²⁵

Variables

Patients aged ≥ 85 years and patients aged 18–35 years, all treated with ECT between 2008 and 2022, whose treatment was identified in Q-ECT and who were admitted as inpatients for moderate to severe depression while receiving the ECT treatment, were initially included in the study (ICD-10 diagnoses: F32.1, F32.2-F32.3, F33.1-F33.3). In addition, a second comparison group of patients aged ≥ 85 years, who were admitted for depression but did not receive ECT, was identified from the NPR.

In Q-ECT, the treatment effect was evaluated using the Clinical Global Impression Improvement rating (CGI-I), which was assessed by the treating physician within 1 week after the end of treatment. The clinical response was divided into two categories: 1) Improved = much to very much improved; and 2) Not improved = unchanged, minimally improved, minimally impaired, and much to very much impaired.³⁰ Patients who lacked a CGI-I assessment were excluded, resulting in zero missing data from this variable in the analyses. Illness severity at baseline, to be used in the propensity score matching procedure, was assessed using the Clinical Global Impression–Severity CGI-S,²⁶ again administered by the treating physician within 1 week from start of the ECT. The Montgomery–Åsberg Depression Rating Scale–Self Assessment (MADRS-S), a nine item depression scale in which each item ranges from 0 to 6 (max score, 54), was used to assess remission from depression. Remission was defined as having a total MADRS_S score of 10 or less post ECT.³¹ Information about previous ECT, ECT parameters, clinician-reported adverse events from ECT, age at treatment, and sex were also derived from the Q-ECT. The clinicians who administered the ECT were expected to report information about the treatment via an online form. Any adverse events that occurred during treatment could be reported via a list for more common

adverse events, or in free form format for uncommon adverse events. Mild adverse events are likely under-reported.

Potential comorbid conditions were derived from the Charlson Comorbidity Index (CCI), which consists of somatic diseases which are associated with mortality and often occurs in older people.^{32,33} More specifically, the following diseases were included: rheumatic arthritis, peripheral vascular disease, metastatic cancer, chronic obstructive pulmonary disease, congestive heart failure, gastrointestinal ulcers, myocardial infarction, severe liver disease, severe kidney disease, hemiplegia, diabetes, dementia, and cerebrovascular disease. The diagnoses were attained from the NPR. The CCI was calculated based on weights adapted for Swedish register data.³³

Severe adverse events requiring hospitalization within 7 days from discharge were derived from the NPR, and date and cause of death within 7 days from discharge were obtained from the CDR. The main cause of hospitalization was reported by the treating medical doctor, as was the cause of death.

Statistical Analyses

All statistical analyses were performed in SAS, v 9.4. The case group of oldest-old patients receiving ECT for depression is from now on called the ‘Oldest-old ECT group’. The comparison groups were propensity score matched at a 1:1 ratio using an “optimal” matching algorithm and a maximum 0.2 caliper width in propensity score logit.³⁴ The first group of young patients receiving ECT for depression, from now on called the “Young ECT group,” was propensity score matched on depression severity (moderate, severe without psychotic features and severe with psychotic features), sex, whether the patient had had previous ECT according to the Q-ECT, electrode placement, and illness level at baseline according to the CGI-S. The second group of oldest-old patients being inpatient treated for depression although not with ECT, henceforth labeled the “Oldest-old non-ECT group,” was propensity score matched on depression severity (moderate, severe without psychotic features and severe with psychotic features), sex, age at treatment, as well as on comorbid rheumatic arthritis, peripheral vascular disease, metastatic cancer, chronic obstructive pulmonary disease, congestive heart failure, gastrointestinal ulcers,

myocardial infarction, severe liver disease, severe kidney disease, hemiplegia, diabetes, dementia, and cerebrovascular disease.

Means, standard deviations, and proportions were calculated for key variables (for each group separately). Conditional binomial logistic regression models were used to assess differences between the case group and the respective comparison group regarding response rates, frequency of adverse events, hospitalizations, and death within 1 week from discharge. If the prevalence of the outcome was less than 10, differences were assessed with Fisher's exact tests. To assess the influence of prognostic factors on treatment outcomes in the oldest-old, a binomial logistic regression analysis was used. Specifically Firth's bias-reduced logistic regression was utilized to handle high separation between predictors and the outcome. The analyses were both univariate and multivariate. Odds ratios for improvement are presented along with a 95% confidence interval. A p-value of <0.05 was considered statistically significant.

RESULTS

Participants

A total of 522 patients aged between 85 and 99 years old who received ECT for depression were identified (the Oldest-old ECT group). Of these, 377 (72.2 %) were women. In addition, 522 patients aged 18–35 who received ECT for depression (Young ECT group), as well as 522 patients aged between 85 and 96, who were inpatient treated for depression with methods other than ECT (Oldest-old non-ECT group), were identified. The characteristics of the patients in the three groups are presented in Table 1.

Severe depressive disorder without psychotic symptoms was the most common diagnosis in all three groups. Even though the groups were propensity score matched, the Oldest-old ECT group and the Young ECT group included more cases of severe psychotic depression than the Oldest-old non-ECT group, indicating that the groups that received ECT were of a more severely

TABLE 1. Characteristics of Patients Aged ≥ 85 Treated With ECT, Patients Aged 18–35 Treated With ECT, and Patients Aged ≥ 85 Receiving Treatments Other Than ECT, for Depression

	Oldest-Old ECT Group	Young ECT Group	Oldest-Old Non-ECT Group
<i>Group size (n)</i>	522	522	522
<i>Age, M (SD/Range)</i>	87.2 (2.4/85–99)	28.0 (4.6/18–35)	87.1(2.3/85–96)
<i>Sex, n (%)</i>			
Female	377 (72.2)	366 (70.1)	375 (71.8)
Male	145 (27.8)	156 (29.9)	147 (28.2)
<i>Diagnosis, n (%)</i>			
MDD, moderate F32.1/F33.1	171 (32.8)	171 (32.8)	180 (34.5)
MDD, severe without psychotic features F32.2/F33.2	263 (50.4)	263 (50.4)	259 (49.6)
MDD, severe with psychotic features F32.3/F33.3	88 (16.9)	88 (16.9)	83 (15.9)
<i>ECT treatment prior to index ECT, n (%)</i>			
Yes	395 (75.7)	394 (75.5)	^a
No	127 (24.3)	128 (24.5)	^a
<i>Co-morbidity, n (%)</i>			
Rheumatic arthritis	64 (12.3)	9 (1.7)	62 (11.9)
Peripheral vascular disease	14 (2.7)	1 (0.2)	18 (3.5)
Metastatic cancer	4 (0.8)	0 (0.0)	6 (1.2)
COPD	6 (1.2)	3 (0.6)	6 (1.2)
Congestive heart failure	24 (4.6)	0 (0.0)	22 (4.2)
Peptic ulcer disease	14 (2.7)	2 (0.4)	11 (2.1)
Myocardial infarction	41 (7.9)	0 (0.0)	35 (6.7)
Severe liver disease	2 (0.4)	1 (0.2)	0 (0.0)
Severe kidney disease	12 (2.3)	2 (0.4)	7 (1.3)
Hemiplegia	3 (0.6)	1 (0.2)	4 (0.8)
Diabetes	22 (4.2)	11 (2.1)	19 (3.6)
Dementia	16 (3.1)	0 (0.0)	14 (2.7)
Cerebrovascular disease	64 (12.3)	3 (0.6)	68 (13.0)

MDD: major depressive disorder; ECT: electroconvulsive therapy; COPD: chronic obstructive pulmonary disease.

^a Precise information about prior ECT was available only from the Q-ECT, from which the ECT groups were derived, and therefore was not available for the Oldest-old non-ECT group.

TABLE 2. ECT-related parameters in the Oldest-old and Young ECT groups

	Oldest-old ECT group	Young ECT group	$\chi^2(\text{df})/\text{t}(\text{df})$	P
Response, yes (%)	428 (82.0)	352 (67.4)	29.18 (1)*	<.001
Remission, yes (%)	130 (53.3)	79 (27.4)	21.88 (1)*	<.001
Electrode placement			2.02 (3)**	.568
Unilateral, n (%)	473 (91.1)	464 (89.2)		
Bilateral, n (%)	42 (8.1)	48 (9.2)		
Other, n (%)	4 (0.8)	8 (1.5)		
ECT sessions per series, Med. (IQR)	6.0 (2.0)	8.0 (3.0)	3.92 (1042)***	<.001
Seizure time, s, Med. (IQR)	49.0 (30.0)	47.0 (30.0)	2.97 (815)***	.003
Pulse width, ms, Med. (IQR)	0.50 (0.05)	0.50 (0)	6.44 (842)***	<.001
Frequency, Hz, Med. (IQR)	70.0 (20.0)	50.0 (30.0)	17.22 (839)***	<.001
Duration, s, Med. (IQR)	6.9 (2.0)	6.0 (2.4)	4.12 (841)***	<.001
Current, mA, Med. (IQR)	800.0 (100.0)	800.0 (100.0)	1.20 (841)***	.232
Charging dose, mC, Med. (IQR)	427.5 (194.0)	251.0 (143.0)	19.35 (841)***	<.001

Note. Differences between groups were assessed using t-tests (continuous variables) and the chi square test (categorical variables). Med., Median. IQR, interquartile range.

*The comparison was assessed by logistic binomial regression.

**The comparison was assessed by a chi square test.

***The comparisons were assessed with t-tests.

psychiatrically ill population than the non-ECT group. Regarding comorbidities, rheumatic arthritis and cerebrovascular disease were the most common diagnoses among the oldest-old, while diabetes was the most common diagnosis in the young age group.

Response Rates, Remission Rates, and ECT Parameters

Significantly more of the 522 patients in the Oldest-old ECT group (428, 82.0%) responded to ECT than in the Young ECT group (352, 67.4%; OR 2.38, 1.74–3.26), and significantly more patients in the oldest-old group remitted (130, 53.3%) than in their younger counterpart group (79, 27.4%; OR 4.23, 2.31–7.74).

Patients in the Oldest-old ECT group received slightly fewer ECT sessions per series, on average, than those in the Young ECT group. Electrode placement was unilateral in a majority of cases. The average seizure time was similar across groups.

The characteristics of ECT setup were delineated by several key parameters. On a general level, the Oldest-old ECT group was characterized by higher ECT parameter values than the Young ECT group. Table 2 provides a comprehensive presentation of ECT-related parameters across both age groups.

Factors Prognostic for Responses to ECT and Remission From Depression

Associations between potential prognostic factors, treatment response, and remission from

depression in the Oldest-old ECT group are summarized in Table 3. The only prognostic factor with a significant association with the outcome was psychotic depression, which was associated with a higher likelihood of both a response to ECT and remission from depression.

Clinician-Reported Adverse Events Associated With ECT

Among the patients in the Oldest-old ECT group, 103 (19.7 %) experienced adverse events during treatment with ECT (as reported by the treating clinician). These adverse events were not necessarily caused by ECT. One-hundred-and-twenty-one (23.2 %) patients in the Young ECT group had recorded adverse events, a nonsignificant difference compared to the Oldest-old ECT group. The predominant adverse events reported among the oldest-old were memory impairment, confusion, cardiovascular complications (including alterations in blood pressure and arrhythmias), and headaches (Table 4). In the younger age group, memory impairment was the most prevalent adverse event, followed by headaches. Eight patients in this group had confusion, and four exhibited cardiovascular complications.

Note: Clinicians may have reported more than one adverse event per patient. Adverse events emerging during ECT may not necessarily be caused by ECT.

Severe Adverse Events Requiring Hospitalization or Causing Death of Oldest-Old Patients During or Following Treatment for Depression, and of Young Patients Receiving ECT

Among the Oldest-old ECT group, two patients died during or within 1 week from finishing ECT, and 30 were hospitalized within 1 week from discharge. By contrast, a higher proportion (nine patients) in the Oldest-old non-ECT group (OR 4.56, two-sided Fisher's exact test $p = 0.06$) died, and 72 were hospitalized within 1 week from discharge (OR 2.68, 95% CI 1.69–4.24, Wald χ^2 (df) = 17.69 (1), $p < 0.01$), meaning that although the groups did not differ significantly regarding mortality, significantly more patients in the non-ECT group were hospitalized again within 1 week from discharge. Among the Young ECT group, none died and 23 were hospitalized within 1 week from discharge. The reasons for hospitalizations or death are summarized in [Table 5](#), which shows that no cause of hospitalization or death that may be linked to ECT stood out in the Oldest-old ECT group compared with the Oldest-old non-ECT group, nor in the Oldest-old ECT group compared with the Young ECT group. Conditions such as disorientation and mild cognitive disorder might be linked to ECT, but we found that they were more common in the non-ECT group. If anything, the Oldest-old non-ECT group reported a higher prevalence of hospitalizations and deaths for those with both psychiatric and somatic diagnoses compared with the Oldest-old ECT group, although the number of hospitalizations for a specific diagnosis did not differ significantly.

This population-based register study aimed to assess response rates, prognostic factors, and adverse events in 522 oldest-old patients (85–99 years old) treated with ECT for depression. We found that 81.2% of the oldest-old patients responded to ECT compared with 67.4% of a young comparison group of 18–35 year olds, a statistically significant difference. In addition, significantly oldest-old patients remitted from depression following ECT (53.3%) than

TABLE 4. Prevalence of the Most Common Clinician-Reported Adverse Events During Treatment With ECT in Patients Aged ≥ 85 (Oldest-Old ECT Group) or 18–35 Years (Young ECT Group)

	Oldest-Old ECT Group, n (% of Age Group)	Young ECT Group, n (% of Age Group)	Odds Ratio (95% CI)	Wald χ^2 (df), p
Adverse events, total n (%)	103 (19.7)	121 (23.2)	0.82 (0.61–1.10)	1.79 (1), 0.180
Specific adverse events				
Memory impairment	39 (7.5 %)	69 (13.2 %)		
Confusion	23 (4.4 %)	8 (1.5 %)		
Cardiovascular complications	23 (4.4 %)	4 (0.8 %)		
Headache	8 (1.5 %)	39 (7.5 %)		
Nausea/Vomiting	6 (1.1 %)	8 (1.5 %)		
Muscle pain	3 (0.6 %)	7 (1.3 %)		
Suicide attempt	0 (0.0 %)	6 (1.1 %)		

younger patients (27.4 %). These findings are in line with previous studies in the field, which report response rates to ECT as high as 80%–97.5% in older adults.^{19,35,36} A recent study found response and remission rates after ECT in old patients were higher than in younger patients,¹⁷ and similar results were found by two earlier studies.^{37,38} The exact proportions may vary due to different study samples and different ways of defining the outcome.

In this study, the only prognostic factor with a significant association with response to ECT and remission from depression was severe depression with

psychotic features; neither comorbid somatic diagnoses nor ECT parameters were associated significantly with these outcomes. This result is consistent with previous studies indicating that ECT is especially effective in patients with psychotic symptoms,^{39–41} and particularly in older patients.^{15,42} Comorbid somatic diagnoses were not associated with mortality or acute somatic events in a previous study including almost 7,000 patients who received ECT,⁴³ and another study indicated that also in old patients, somatic comorbidities were not associated with side effects or complications from ECT.⁴⁴ The current

TABLE 5. Causes of Hospitalization or Death (Within 1 Week From Discharge) in Oldest-Old Patients Treated for Depression Either With or Without ECT, as Well as in Young Patients (Aged 18–35) Who Received ECT

	Young ECT Group, n (%)	Oldest-Old ECT Group, n (%)	Oldest-Old Non-ECT Group, n (%)
Total number of hospitalizations, n (%)	23 (4.4)	30 (5.7)	72 (13.8)
Cause of hospitalization			
Infectious diseases	0 (0)	1 (0.2)	2 (0.4)
Cancer	0 (0)	0 (0.0)	2 (0.4)
Organic mental disorders ^a	0 (0)	1 (0.2)	7 (1.3)
Depression relapse	10 (2.2)	7 (1.3)	14 (2.7)
Other psychiatric condition	7 (1.3)	0 (0)	0 (0)
Dissociative stupor	0 (0)	1 (0.2)	0 (0.0)
Cardiovascular disease	0 (0)	5 (1.0)	8 (1.5)
Cerebrovascular disease	0 (0)	1 (0.2)	2 (0.4)
Influenza or pneumonia	0 (0)	4 (0.8)	6 (1.1)
Urinary disease	0 (0)	2 (0.4)	1 (0.2)
Syncope and collapse	0 (0)	1 (0.2)	0 (0.0)
Intestinal diseases	0 (0)	0 (0.0)	5 (1.0)
Musculoskeletal diseases	0 (0)	0 (0.0)	5 (1.0)
Traumatic injury	1 (0.2)	4 (0.8)	6 (1.1)
Pregnancy-related	2 (0.4)	0 (0)	0 (0)
Poisoning	5 (1.0)	0 (0)	0 (0)
Other	0 (0)	3 (0.6)	6 (1.1)
Total number of deaths, n(%)	0 (0)	2 (0.4)	9 (1.7)
Cause of death			
Heart failure	0 (0)	1 (0.2)	2 (0.4)
Infection	0 (0)	0 (0.0)	3 (0.6)
Asphyxiation	0 (0)	0 (0.0)	1 (0.2)
Unspecified	0 (0)	1 (0.2)	3 (0.6)

^a Including dementia, delirium, disorientation and mild cognitive disorder.

results of the study extend these findings by showing that somatic comorbidities in oldest-old patients do not seem to influence response and remission rates in either direction. Although it seems that there is no association between somatic comorbidities and response to ECT, it may be that the oldest-old patients are a select group of relatively healthy people, having survived to such an advanced age. Indeed, the prevalence of age-related comorbidities in this sample is relatively low for an older population with severe depression.

Regarding clinician-reported adverse events associated with ECT, slightly fewer of the oldest-old patients had at least one adverse event reported (19.7%) compared with the young comparison group (23.2%), although the adverse event profiles of these groups differed. Notably, more patients in the younger age group than in the oldest-old age group had post-treatment memory impairment and headaches reported. Conversely, confusion and cardiovascular complications were documented more frequently among the oldest-old. It is worth considering that the heightened incidence of such adverse events among the oldest-old patients may be due to individuals in this age category already experiencing them before treatment; however, it is also possible that older patients are more susceptible to these particular adverse events, indicating a need for increased caution when administering ECT to this demographic. An interesting result was that more young people seemed to experience an impact on memory after ECT. This has also been observed in previous studies regarding both subjective memory impairment⁴⁵ and performance on neuropsychological tests.⁴⁶ Conversely, a recent study observed a lesser decline in cognitive function in older than in younger patients,¹⁷ and other studies have even observed an improvement in memory in older patients following ECT.^{47,48} One potential explanation for this could be that younger people tend to have a better memory than older people before ECT, and that even a small decrease in memory function in young patients could be significant; however, the difference in objective tests from previous studies indicate that this cannot explain the difference completely.

The oldest-old who received ECT had significantly less severe adverse events requiring hospitalization than their oldest-old counterparts who were not treated with ECT. This is despite the fact that they on

a group level appeared to have similar profiles regarding comorbid conditions. Although we must be wary of drawing too far-fetched conclusions based on the results of a single study, it appears that treating an oldest-old patient suffering from depression with ECT is not associated with a heightened risk of adverse events compared with treating the patient with an alternative treatment option.

This study has some limitations that warrant consideration. First, the information derived from national registers is usually limited, although the sample sizes are large. For example, assessment of response to ECT relied on clinical evaluation using the CGI-I scale, which is a crude measure. We also lacked information about which physician made each specific assessment, and what the inter-rater reliability across ratings were. In addition, the registers lack potentially relevant variables such as cognitive function, psychotropic comedication, treatment resistance, or catatonic features, which we therefore could not include in the propensity score matching; in addition, and registry data provide no indication of the degree of nonreported. Also, variables such as the clinician-reported adverse events may have been underreported, and hospitalization within 7 days from ECT is a rather broad measure. Furthermore, a notable limitation is concurrent utilization of alternative treatment methods such as antidepressant drug treatment alongside ECT in a substantial number of patients. This co-treatment could potentially introduce confounding variables that were not explicitly addressed in this study. Importantly, we also did not have any information about what treatment(s) the patients in the Oldest-old non-ECT group may have received. Another important limitation is the risk of selection bias inherent in register-based studies. For instance, even though the Oldest-old non-ECT group was propensity score matched, the group may still have constituted a generally more physically healthy cohort at the time of treatment, in ways the available variables failed to capture. This, in turn, may have explained the higher prevalence of hospitalizations in the non-ECT group. Conversely, the ECT group may have constituted a more severely mentally ill population than the non-ECT group. It is also adequate to highlight the strengths of this study. Notably, the study benefits from a large and diverse patient cohort from a real-world setting. To the best of our knowledge, this is by far the largest study to date

assessing response and remission rates, as well as adverse events, in oldest-old patients treated with ECT. Moreover, evaluation of outcomes was standardized and applied consistently to all patients, ensuring uniform assessment practices. We also included both a clinician-rated and a patient-reported measure to assess outcomes of ECT to increase the validity of the results. Furthermore, the standardized nature of ECT procedures across hospitals in Sweden contributes to the study's internal consistency, as well as the comprehensive coverage provided by the quality register, encompassing almost all patients in Sweden who have received ECT treatment.

In conclusion, this study revealed high response rates and remission rates from depression in the oldest-old patients (85+ years old) treated with ECT, significantly higher than those in a younger control group. The oldest-old patients also reported significantly fewer clinician-reported adverse events than their younger counterparts, including less subjective memory impairment, although the nature of the side effects differed. Compared with oldest-old patients who did not receive ECT, those who did so experienced fewer adverse events requiring hospitalization in close proximity to discharge. Although this study is limited by the risk of residual confounding, and further research is warranted, our results (along with current evidence) indicate that ECT should be considered an important and safe complement to pharmacological treatment of depression in the oldest-old, particularly for treating severe depression with psychotic features.

AUTHOR CONTRIBUTIONS

Tor Arnison: Writing—original draft, Data curation, Formal analysis, Methodology, Software, Conceptualization,

Funding acquisition. Alina Eriksson: Writing—original draft, Formal Analysis, Conceptualization. Axel Nordenskjöld: Writing—review and editing, Supervision, Validation, Resources, Project Administration, Funding Acquisition, Data Curation, Software, Conceptualization.

ETHICAL CONSIDERATIONS

The study was approved by the Regional ethics vetting board of Uppsala (registration number 2014/174), and by the Swedish Ethical Review Authority (2021-03815). Patients could choose not to have data on ECT included in the Swedish National Quality Registry.

DATA STATEMENT

The data that support the findings of this study are available from Axel Nordenskjöld upon reasonable request.

DISCLOSURES

The authors report no conflicts with any product mentioned or concept discussed in this article.

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